

Task 1P

Solve the following set of problems using Python and submit the code file with extension .ipynb.

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1. Read "[Sensor readings, with temperature, light, humidity every 5 minutes at 8 locations \(trial, 2014 to 2015\)](#)" and print the feature name with numbers of missing entries.
 2. Fill in the missing entries. For filling any feature, you can use either the mean or median value of the feature values from observed entries. Explain the reason behind your choice and print replacement value of each feature.
 3. Use a histogram to show the distribution of the variable "location". Explain the distribution and show how many locations are there in total.
 4. Encode the categorical variable "location" using an appropriate encoding approach and display the encoded values. Justify your selection of encoding approaches.
 5. Apply the min-max scaling on features "[boardtype, boardid ,temp_max, temp_min, temp_avg,light_max, light_min,light_avg, humidity_min, humidity_max, humidity_avg]" . Plot distribution of first six features before and after scaling. Is there any difference? Please explain.

Assessment feedback

The results with comments will be released within 5 business days from the due date.

Referencing

You must correctly use the IEEE method in this assessment. See the [referencing guide](#).

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